

Project Progress Presentation-I
on

Automated Mobile Robotic Arm for Cold Storage Operations



Presented By

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Project Guide

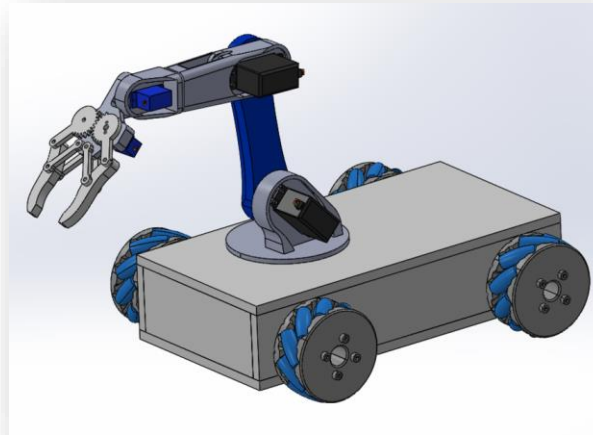
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Updated Literature Review



Relevant Technologies:

- Robotics and automation are already widely used in industries like logistics and cold storage. However, most of these systems are either stationary or have limited mobility. Technologies like automated conveyor belts, robotic forklifts, and stationary robotic arms are commonly used to improve efficiency in cold environments.

Gaps in Knowledge:

- Although many studies focus on robots working in challenging environments, there is limited research on mobile robotic systems specifically designed to operate in cold storage. Most robotic systems are not built to handle the combination of mobility, precision, and durability required in such low-temperature environments.

Suggestions from Previous Presentation

- Suggestion 1:** Change title to better reflect the project's scope.

Compliance: Updated the title to “Automated Mobile Robotic Arm for Cold Storage Operations.”

- Suggestion 2:** Consider how the robotic arm will sustain operation in a cold storage environment.

Compliance: Selected the MG996R servo motor (operating range -10°C to +60°C) and Samsung Cell Premium Li-Ion 7.4V 2600mAh battery (operating range -20°C to +60°C).

Progress – Servo Motor Selection



- Final Selection:** we selected the MG996R servo motor, which is capable of functioning within a temperature range of -10°C to $+60^{\circ}\text{C}$, making it suitable for cold storage conditions.
- Performance:** The MG996R motor is known for providing sufficient torque and precision, which is crucial for the accurate movement of the robotic arm. It has a maximum torque of 11 kg/cm, ensuring the arm can handle moderate loads with high accuracy, even in low-temperature environments.

Progress – Battery Selection



- After evaluating various battery options, we selected the Samsung Cell Premium Li-Ion 7.4V 2600mAh, suitable for cold environments (-20°C to 60°C).
- Ensures prolonged operation of the robotic arm in cold conditions.

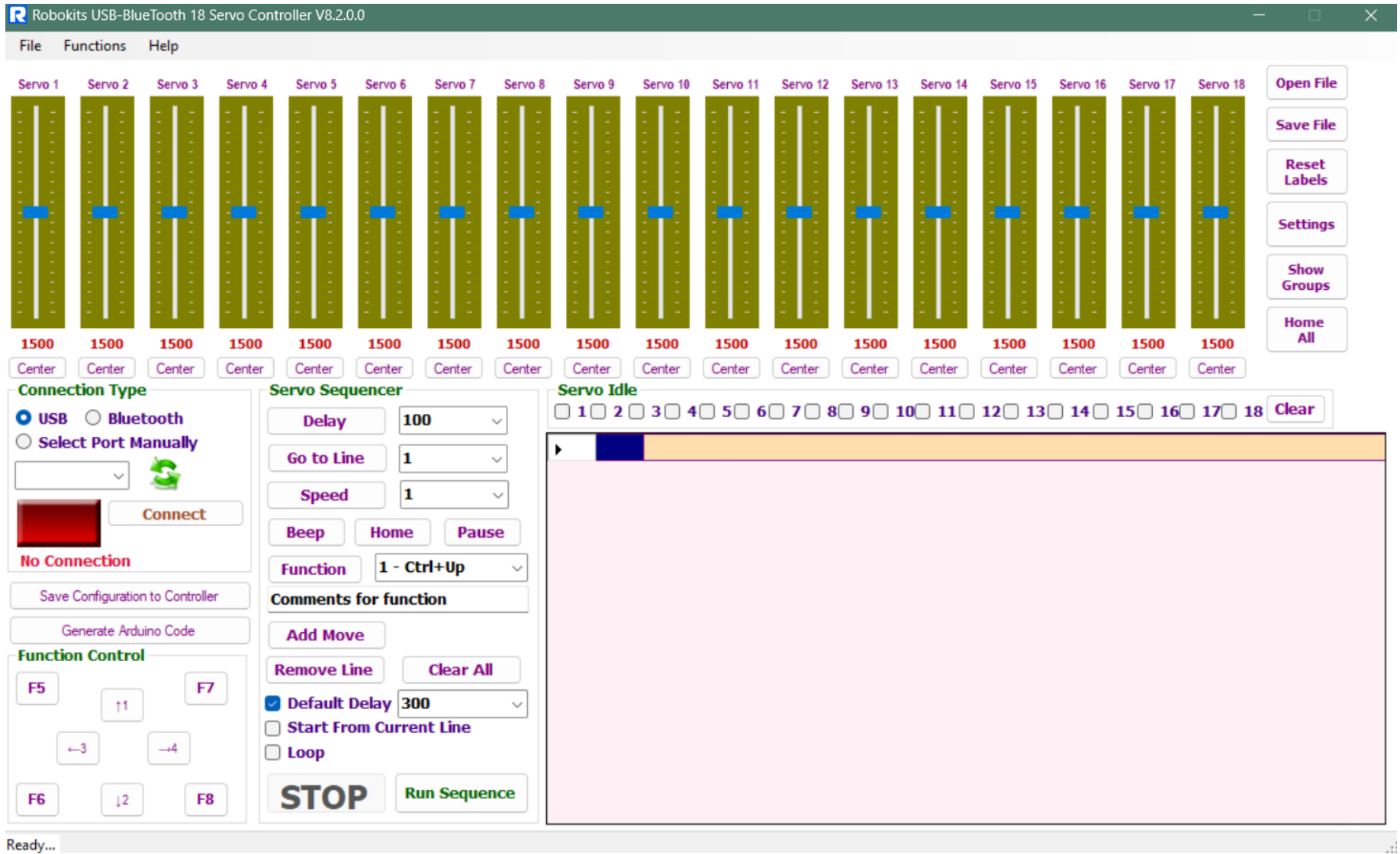
Progress – Servo Controller



or



- Previous Challenges:** In earlier projects, coding for precise and accurate movement of the robotic arm was challenging due to the complexity of managing motor positions with traditional methods.
- Solution Found:** After extensive research, we found a servo controller that can control the MG996R servo motors through a PC or laptop using Bluetooth technology. This controller allows for real-time adjustments and more precise control of the robotic arm's movements, simplifying the programming and ensuring smoother operation.



Project cost

Name	Quantity	Price Per Quantity	Total Price	Order
Control Board	1	2596	2596	yes
Servo Motor MG 996r	3	335	1005	yes
micro Servo Motor	1	149	149	
Stepper	2	1799	3598	
Stepper Driver	4	209	836	
battary	1	587	587	yes
Filament	1	1800-2000	2000	
M6 Self Locking Nut	1	0-10	10	
M6×25 Screw	1	0-10	10	
M3 self-locking nut	2	0-15	30	
M3×20 Screws	2	0-15	30	
M3×10 Hex Recessed Head Screw	1	0-10	10	
M4 Self-locking Nut	9	0-15	135	
M4×40 Screw	1	0-10	10	
M4×30 Screw	1	0-10	10	
M4×60 mm Threaded Rod	1	0-10	10	
M4×32 mm Threaded Rod	1	0-15	15	
606zz Bearing	1	0-15	15	
M4 Washers	15	0-15	225	
M4×20	5	0-15	75	
Dia 6 mm Ball Spheres	25	0-20	500	
		Total Cost	11856	

Cold Environment Simulation and Testing Plan

- Next steps involve testing the system in a simulated cold storage environment.
- Planning to measure battery life, servo motor performance, and overall system stability at low temperatures.

Future Work

- Complete testing of the robotic arm in a cold storage environment.
- Fine-tune control systems and navigation.
- Explore further automation and AI integration for path planning.

THANK YOU